

JEE MAIN REVISION MASTER: BIOMOLECULES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 27 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Carbohydrates (Glucose, Fructose & Glycosidic Linkage)	16 PYQs	Carbohydrates
High	Amino Acids, Peptide Bonds & Nucleic Acids	11 PYQs	Proteins & DNA

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Carbohydrates (Glucose, Fructose & Glycosidic Linkage)	$\text{Sucrose (non-reducing)} \xrightarrow{\text{hydrolysis}} \text{D-(+)-glucose} + \text{D-(-)-fructose} \quad \text{Invert sugar, levorotatory}$
Amino Acids, Peptide Bonds & Nucleic Acids	$\text{Peptide linkage: } -\text{CO-NH}-, \quad \text{Zwitterion: } \text{H}_3\text{N}^+-\text{CHR}-\text{COO}^-, \quad \text{DNA: } \text{A}=\text{T}, \text{G} \equiv \text{C}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Carbohydrates (Glucose, Fructose & Glycosidic Linkage) (Very High Yield — 16 PYQs)

Core Principle: Structure of D-glucose and D-fructose, cyclic hemiacetals, anomeric carbons (α/β), reducing vs non-reducing sugars, and invert sugar.

■ **Caution / JEE Pitfall:** Sucrose is a NON-REDUCING sugar because both reducing groups (hemiacetal -OH of glucose and hemiketal -OH of fructose) are tied in glycosidic linkage.

■ **Speed Trick:** Hydrolysis of dextrorotatory sucrose produces a levorotatory mixture because fructose has a larger specific rotation (-92.4°) than glucose ($+52.5^\circ$).

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Consider the following statements: Statement-I: All monosaccharides are reducing sugars. Statement-II: Sucrose can reduce ammoniacal silver ..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Match List-I with List-II List-I (Vitamins) List-II (Deficiency Diseases) (a) B₁ (i) Convulsions (b) B₂ (ii) RBC def..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Which examples of carbohydrates exhibit α -link, (α -glycosidic link) in their structure?..." [Ans: N/A]

3.2 Amino Acids, Peptide Bonds & Nucleic Acids (High Yield — 11 PYQs)

Core Principle: Essential amino acids, zwitterionic dipolar character, protein denaturation, double-helix structure of DNA, and hydrogen-bonded base pairs.

■ **Caution / JEE Pitfall:** Denaturation destroys secondary, tertiary, and quaternary protein structures, but leaves the primary sequence (peptide bonds) INTACT.

■ **Speed Trick:** All naturally occurring α -amino acids are optically active EXCEPT Glycine ($\text{H}_2\text{N}-\text{CH}_2-\text{COOH}$), which is achiral.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Incorrect statement about α -amino acids of proteins among the following is..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "The correct sequence of α -amino acids, hormone, vitamin, carbohydrates respectively is..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "The type of linkage present between nucleotides is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Incorrect statement about α -amino acids of proteins among the following is

- (A) Methionine is an essential amino acid
- (B) Glycine doesn't exhibit enantiomerism
- (C) Glycylalanylglycine has three amide linkages
- (D) Zwitterion of valine exhibits amphoteric behaviour

Correct Answer: None

Solution Analysis: The incorrect statement is: Option C Glycylalanylglycine is a tripeptide made from three amino acid units: Glycine-Alanine-Glycine. In a peptide, two amino acids join by one peptide bond, also called an amide linkage. So, for three amino acid units, the number of peptide bonds is: $3-1=2$. Hence, glycylalanylglycine has only two amide linkages, not three. The other statements are correct: Methionine is an essential amino acid. Glycine does not show enantiomerism because its α -carbon has two hydrogen atoms, so it is not chiral. Valine exists as a zwitterion.

Benchmark Problem 2: KCET 2026 (Q#2)

Consider the following statements: Statement-I: All monosaccharides are reducing sugars. Statement-II: Sucrose can reduce ammoniacal silver nitrate solution. Choose the correct answer from the options given below.

- (A) Both Statement I and Statement II are correct
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is correct but Statement II is incorrect
- (D) Statement I is incorrect but Statement II is correct

Correct Answer: None

Solution Analysis: Statement-I is correct. All monosaccharides are reducing sugars because they have a free aldehydic group or ketonic group, or they can form such a group in solution. So they can reduce reagents like Fehling's solution and Tollens' reagent. Statement-II is incorrect. Sucrose is a non-reducing sugar. In sucrose, the glycosidic linkage is formed between the anomeric carbon of glucose and the anomeric carbon of fructose. So, no free aldehydic or ketonic group is available to reduce ammoniacal silver nitrate solution, also called Tollens' reagent. Hence, the correct answer is: Option C: Statement I